in the mid-1980s, field programmable gate arrays quickly exploded in the telecommunication and networking sector in the '90s. FPGAs continued to be in demand for anything to do with a digital circuit and were welcomed with open arms in the field of consumer tech and automobile manufacturing. FPGAs were the new kid on the block and remain instrumental in fields ranging from medical imaging to radio astronomy and everything in between.

Think of them as integrated circuits that can be re-programmed on-the-fly in a jiffy, great for prototyping semiconductor devices or testing "logic function" before perfecting them for large scale deployment. It has literally helped companies in the electronics manufacturing segment save billions of dollars – at the same time, helping Xilinx (who invented FPGA) mint billions itself. FPGAs are seeing increased application in the development of SoCs (used in latest smartphones and tablets; they're also on the radar of Bitcoin miners trying to generate more virtual currency). FPGA's demand is only going to rise in the near future. Why? Intel (the largest semiconductor manufacturer in the world) and Altera (the second largest producer of FPGAs in the world) recently signed a deal which lets Altera produce insanely better, never before seen FPGAs using Intel's tri-gate transistor 14nm process. So watch out, uber cool FPGAs are coming to change your world forever.

Fuel Cell

A fuel cell uses chemical reactions to convert energy from a fuel such as hydrogen to electricity in the presence of an oxidizing agent. Unlike batteries, a fuel cell can produce energy as long as the raw materials are supplied to it. Speaking of raw materials, apart from hydrogen a fuel cell can also use hydrocarbons such as natural gas and methanol. It is generally believed that Sir William Grove invented the first fuel cell in 1839, while it was a German scientist, Christian Friedrich Schoenbein who first noticed the fuel cell effect. What they were trying to do was reverse electrolysis (a processes by which water is split into its founding blocks – hydrogen and oxygen – by passing electricity through it). Once the process is reversed,